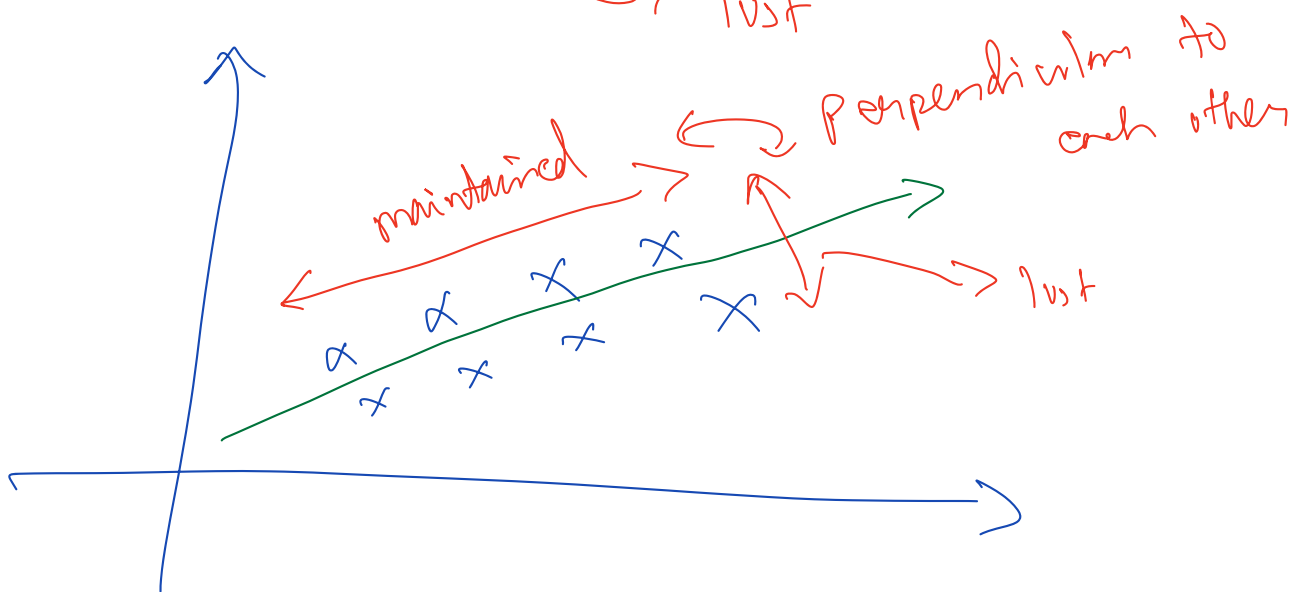
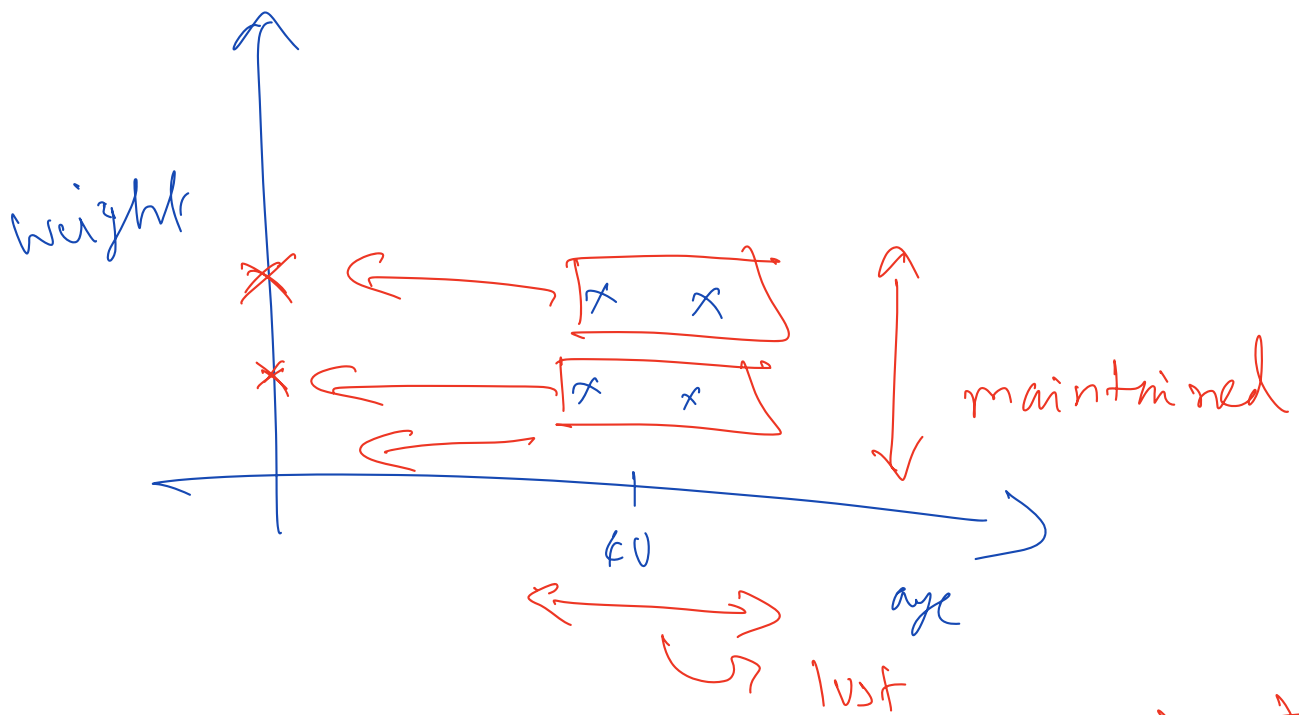


Last Class

- 1) Business Case Intro
- 2) RANSAC & Elliptical Envelope
- 3) Isolation Forests
- 4) One-class SVM (very brief)
- 5) Local Outlier Factor
- 6) Comparison of all methods

Today's class

- 1) Motivation for high-dim viz
- 2) PCA introduction/intuition
- 3) Applying PCA
- 4) Geometrical Obj
- 5) Mathematical proof
- 6) Eigen values and vectors



$$w, h$$

$$PC_1 = \alpha_{11} w + \alpha_{12} h$$

$$PC_2 = \alpha_{21} w + \alpha_{22} h$$

$$\begin{matrix} (w_1, h_1) \\ (w_2, h_2) \end{matrix} \rightarrow \begin{cases} w'_1 = \alpha_{11} w_1 + \alpha_{12} h_1 \\ h'_1 = \alpha_{21} w_1 + \alpha_{22} h_1 \end{cases}$$

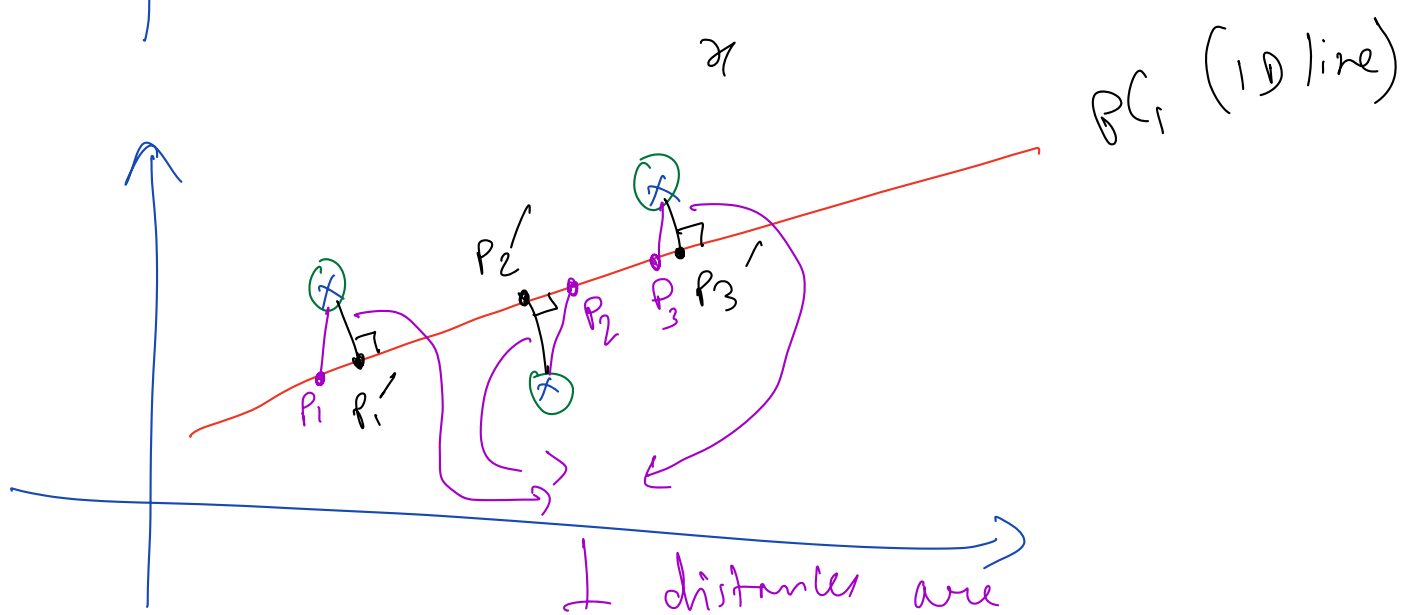
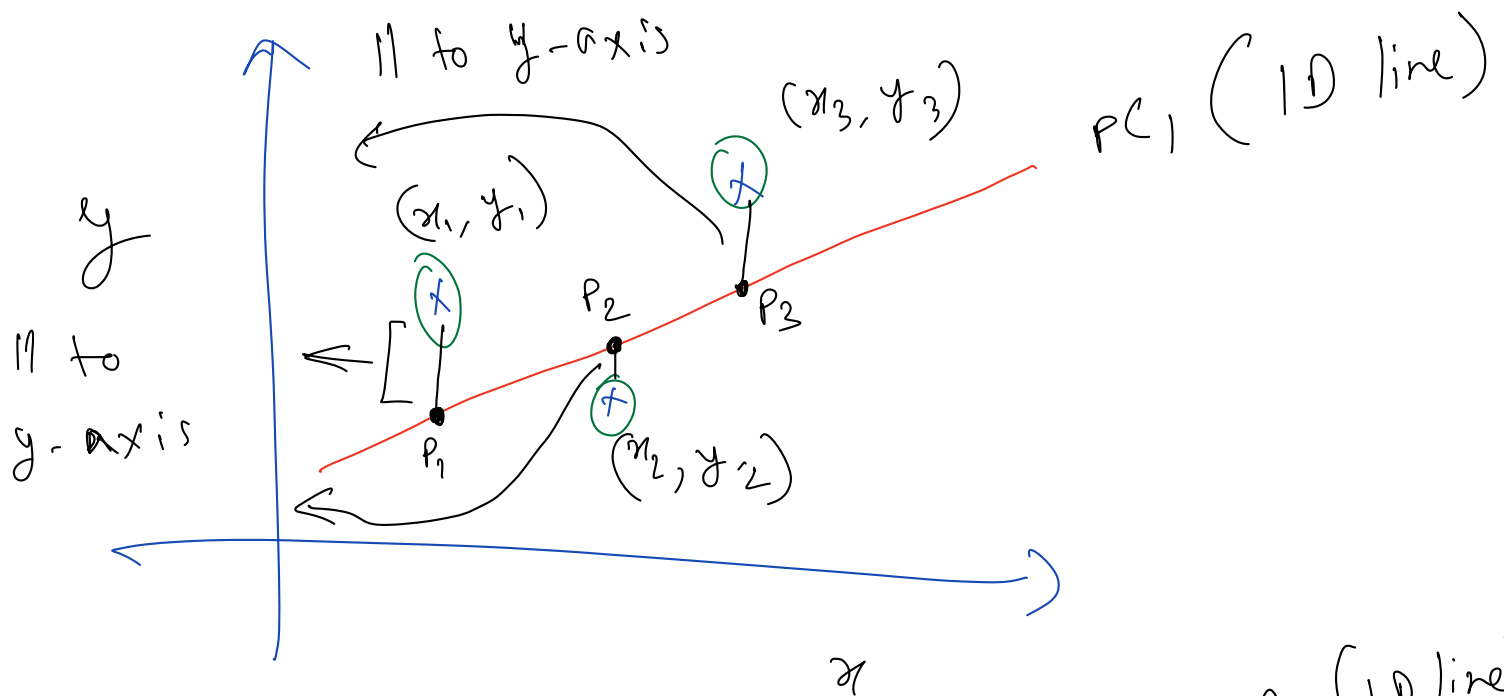
$$\begin{cases} w_2' = \alpha_{11} w_2 + \alpha_{12} h_2 \\ h_2' = \alpha_{21} w_2 + \alpha_{22} h_2 \end{cases}$$

$$\alpha_{11} = 1, \alpha_{12} = 1$$

$w + h$

$$\alpha_{21} = 1, \alpha_{22} = -1$$

$w - h$



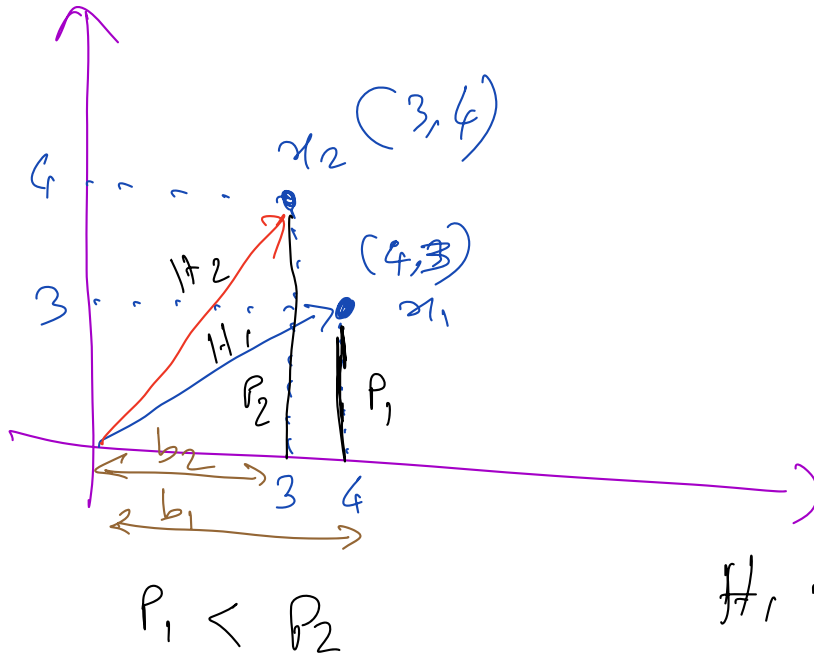
$$P_i = x_i \hat{i} + y_i \hat{j}$$

$$\text{proj } \vec{x}_1 \text{ on } \vec{f}_1 =$$

↓
scalar value

$$\frac{(\vec{x}_1 \cdot \vec{f}_1)}{\|\vec{f}_1\|}$$

→ scalar
→ scalar value



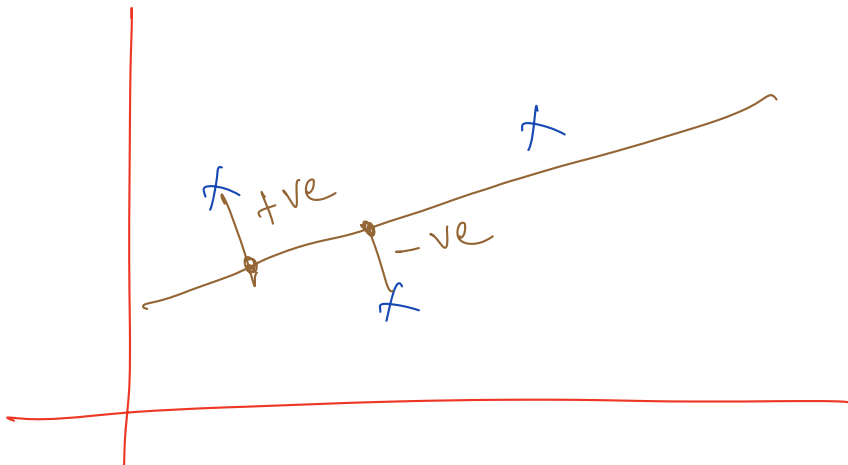
$$h_1 = \sqrt{3^2 + 4^2} = 5$$

$$h_2 = \sqrt{3^2 + 4^2} = 5$$

$$\begin{aligned} p_1 &= 3 & b_1 &= 4 \\ p_2 &= 4 & b_2 &= 3 \\ & & b_2 &< b_1 \end{aligned}$$

$$\frac{\vec{u}}{\|\vec{u}\|}$$

→ unit vector



$$\|\vec{u}\|^2 \rightarrow \text{scalar value}$$

$$\begin{bmatrix} u \\ p \end{bmatrix}_{n \times 1}$$

$$u^T u$$

$$u^T \rightarrow 1 \times n$$

$$= (1 \times n) \times (n \times 1)$$

$$u \rightarrow n \times 1$$

$$(A \ B)^T$$

$$= 1 \times 1 \rightarrow \text{scalar}$$

$$= B^T A^T$$

$$\text{cov} =$$

$$(X - \mu)^T (X - \mu)$$

$\downarrow$

$\downarrow$

0

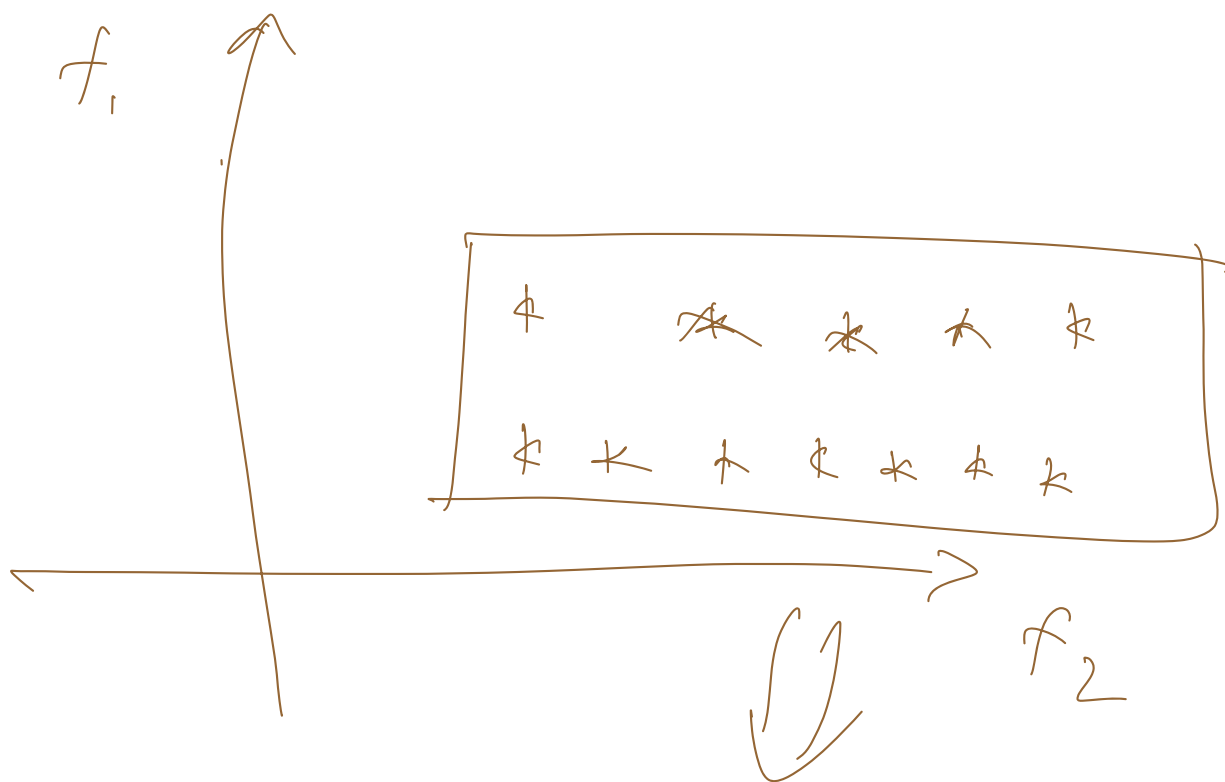
0

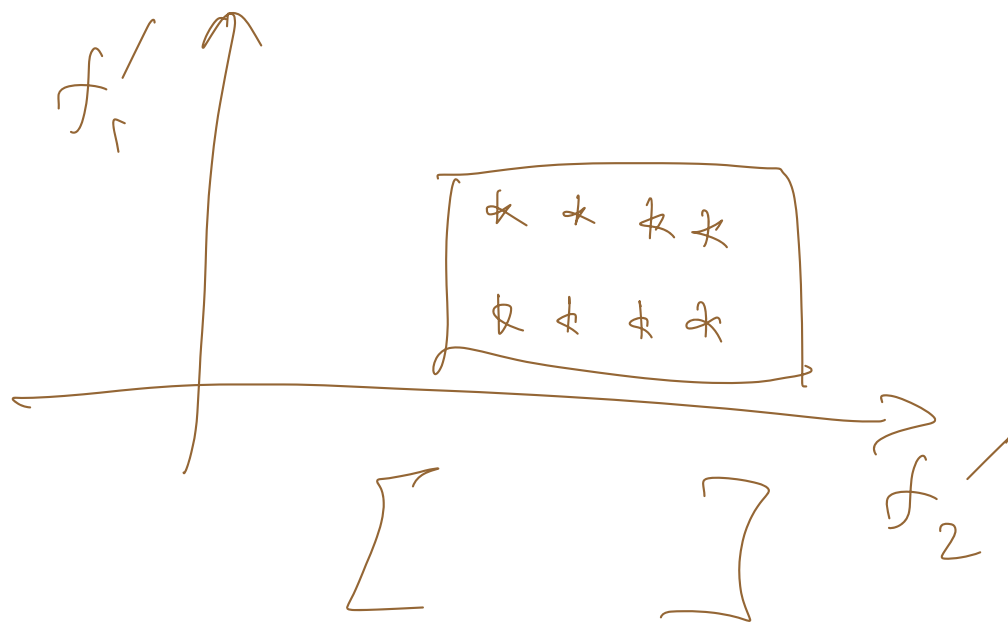
zero mean

$$X^T X$$

$\hookrightarrow$

Cov





$f_1 \rightarrow 1 \text{ to } 10$

$f_2 \rightarrow 1 \text{ to } 100$

100 features

($\rightarrow$) cov matrix

$\rightarrow 100 \times 100$
cov

100 Eigen-values, eigen-vectors

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_{100} \end{bmatrix}$$

$$\begin{matrix} 0.5 \\ 0.4 \\ 0.09 \\ \vdots \\ \end{matrix}$$

Top-k

$$\boxed{\text{top} = 10}$$